

Alessandro Sanvito

Amsterdam, Netherlands

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Experience

Optiver

Amsterdam, Netherlands

SOFTWARE ENGINEER, SYSTEMATIC OPTIONS TRADING

August 2023 - Present

- Expanded the desk beyond index options: built baseline systematic strategies for new product groups from scratch, enabling researchers to rapidly optimize and generate new alpha.
- Selected candidate signals and defined the data-quality and validation criteria each baseline must meet before handoff.
- Built data pipelines processing 10+ TB of instrument-level data with Spark, Polars, and DuckDB, cutting core backtest runtime by 60%; migrated ML model training to AWS.

Mercedes-Benz (AI-SEE Project)

Stuttgart, Germany

AI RESEARCH INTERN AND MASTER THESIS STUDENT

May 2022 - June 2023

- Researched NeRFs and diffusion models for 3D scene reconstruction and human avatar modeling from monocular video in autonomous driving systems.
- Outperformed state-of-the-art baselines in PSNR and SSIM when reconstructing scenes in fog, haze, and limited viewpoints; published first-author at IEEE IV 2024 and co-authored at ICCV 2023.

Education

KTH Royal Institute of Technology

Stockholm, Sweden

MSC. IN ICT INNOVATION, DATA SCIENCE

Sept. 2020 - May 2023

- Final grade: A (Excellent)
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.

Polytechnic University of Milan

Milan, Italy

MSC. IN COMPUTER SCIENCE AND ENGINEERING (DOUBLE DEGREE WITH KTH)

Sept. 2020 - Jul. 2023

- 1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on Twitter's 1B-interaction dataset with the university team under Prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude

Polytechnic University of Milan

Milan, Italy

BSc. IN ENGINEERING OF COMPUTING SYSTEMS

Sept. 2017 - Jul. 2020

- Final grade: 109/110

Publications

2024	HINT: Learning Complete Human Neural Representations from Limited Viewpoints, Sanvito Alessandro (first author), et al.	IEEE IV 2024
2023	ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings , Ramazzina Andrea, et al., incl. Sanvito Alessandro	ICCV 2023
2022	United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources , Maldini Pietro, Sanvito Alessandro , Surricchio Mattia	ACM RecSys 2022
2021	Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment , Carminati Luca, et al., incl. Sanvito Alessandro	ACM RecSys 2021

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, AWS
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest